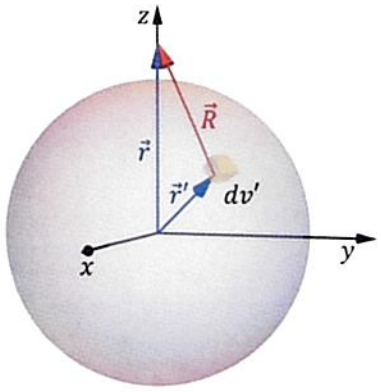
Problema 1.1



$$\bar{E}(z) = \frac{\rho}{4\pi\epsilon_0}\int_{v'}\frac{\bar{R}}{R^3}\,dv'$$

$$\bar{r} = z\,\hat{u}_z$$

$$\bar{r}' = r'\hat{u}_r = r'\left[\cos\theta'\,\hat{u}_z + \operatorname{sen}\theta'\left(\cos\varphi'\,\hat{u}_x + \operatorname{sen}\varphi'\,\hat{u}_y\right)\right]$$

$$\bar{R} = \bar{r} - \bar{r}' = \left(z - r'\cos\theta'\right)\hat{u}_z - r'\operatorname{sen}\theta'\left(\cos\varphi'\,\hat{u}_x + \operatorname{sen}\varphi'\,\hat{u}_y\right)$$

$$R = \left[\left(z - r'\cos\theta'\right)^2 + r'^2\operatorname{sen}^2\theta'\right]^{1/2} = \left(z^2 + r'^2 - 2r'z\cos\theta'\right)^{1/2}$$

Sólo la componente z es no nula, la calculamos:

$$E(z) = \frac{\rho}{4\pi\epsilon_0}\int_0^a\int_0^{\pi}\int_0^{2\pi}\frac{z - r'\cos\theta'}{\left(z^2 + r'^2 - 2r'z\cos\theta'\right)^{3/2}}\,r'^2\operatorname{sen}\theta'\,d\varphi'\,d\theta'\,dr' =$$

$$= \frac{\rho}{2\epsilon_0}\int_0^a\int_0^{\pi}\frac{\left(z - r'\cos\theta'\right)r'^2\operatorname{sen}\theta'}{\left(z^2 + r'^2 - 2r'z\cos\theta'\right)^{3/2}}\,d\theta'\,dr' \overset{①}{=}$$

① $\mu = \cos\theta'$

$d\mu = -\operatorname{sen}\theta'\,d\theta'$

$$= \frac{\rho}{2\epsilon_0}\int_0^a\int_{-1}^{1}\frac{z - r'\mu}{\left(z^2 + r'^2 - 2r'z\mu\right)^{3/2}}\left(-d\mu\right)r'^2\,dr' \overset{②}{=}$$

$$= \frac{\rho}{2\epsilon_0}\int_0^a\frac{r' - z\mu}{z^2\left(r'^2 + z^2 - 2r'z\mu\right)^{1/2}}\Bigg|_{1}^{-1}r'^2\,dr' =$$

$$= \frac{\rho}{2\epsilon_0 z^2}\int_0^a\left(\frac{r' + z}{|r' + z|} - \frac{r' - z}{|r' - z|}\right)r'^2\,dr' \overset{③}{=}$$

③ $z \geq a \geq r' \geq 0$

$|r' - z| = z - r'$

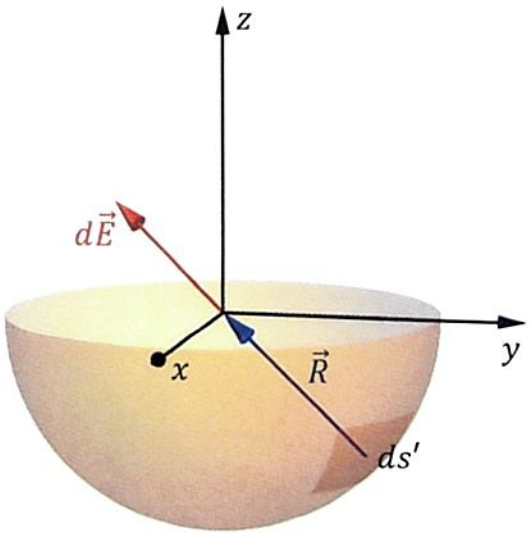
$|r' + z| = z + r'$

$$= \frac{\rho}{\epsilon_0 z^2}\int_0^a r'^2\,dr' = \frac{\rho a^3}{3\epsilon_0 z^2} = \frac{Q}{4\pi\epsilon_0 z^2}$$

$$\bar{E}(z) = \frac{Q}{4\pi\epsilon_0 z^2}\,\hat{u}_z \Rightarrow \bar{E}(\bar{r}) = \frac{Q}{4\pi\epsilon_0 r^2}\,\hat{u}_r$$

② $$\int\frac{ax - b}{\left(a^2 + b^2 - 2abx\right)^{3/2}}\,dx = \frac{a - bx}{b^2\left(a^2 + b^2 - 2abx\right)^{1/2}}$$

Problema 1.12



$\bar{r} = 0$

$\bar{r}' = a\,\hat{u}_r = a\left[\cos\theta'\,\hat{u}_z + \operatorname{sen}\theta'\left(\cos\varphi'\,\hat{u}_x + \operatorname{sen}\varphi'\,\hat{u}_y\right)\right]$

$\bar{R} = -\bar{r}' = -a\left[\cos\theta'\,\hat{u}_z + \operatorname{sen}\theta'\left(\cos\varphi'\,\hat{u}_x + \operatorname{sen}\varphi'\,\hat{u}_y\right)\right]$

$R = a \qquad ds' = a^2 \operatorname{sen}\theta'\, d\theta'\, d\varphi'$

Solo la componente z es no nula.

$$\vec{E}_0 = \frac{\sigma}{4\pi\epsilon_0}\int_{S'}\frac{\bar{R}}{R^3}\,ds' = \frac{\sigma}{4\pi\epsilon_0}\int_{\pi/2}^{\pi}\int_{0}^{2\pi}\frac{-a\cos\theta'}{a^3}\,a^2\operatorname{sen}\theta'\,d\theta'\,d\varphi'\,\hat{u}_z =$$

$$= -\frac{\sigma}{2\epsilon_0}\int_{\pi/2}^{\pi}\cos\theta'\operatorname{sen}\theta'\,d\theta'\,\hat{u}_z =$$

$$= -\frac{\sigma}{2\epsilon_0}\left(\frac{1}{2}\operatorname{sen}^2\theta'\Big|_{\pi/2}^{\pi}\right)\hat{u}_z = -\frac{\sigma}{2\epsilon_0}\left(0-\frac{1}{2}\right)\hat{u}_z =$$

$$= \frac{\sigma}{4\epsilon_0}\,\hat{u}_z = \frac{Q}{8\pi\epsilon_0 a^2}\,\hat{u}_z$$

$$\bar{E}_0 = \frac{Q}{2\pi\epsilon_0 a^2}\,\hat{u}_z$$